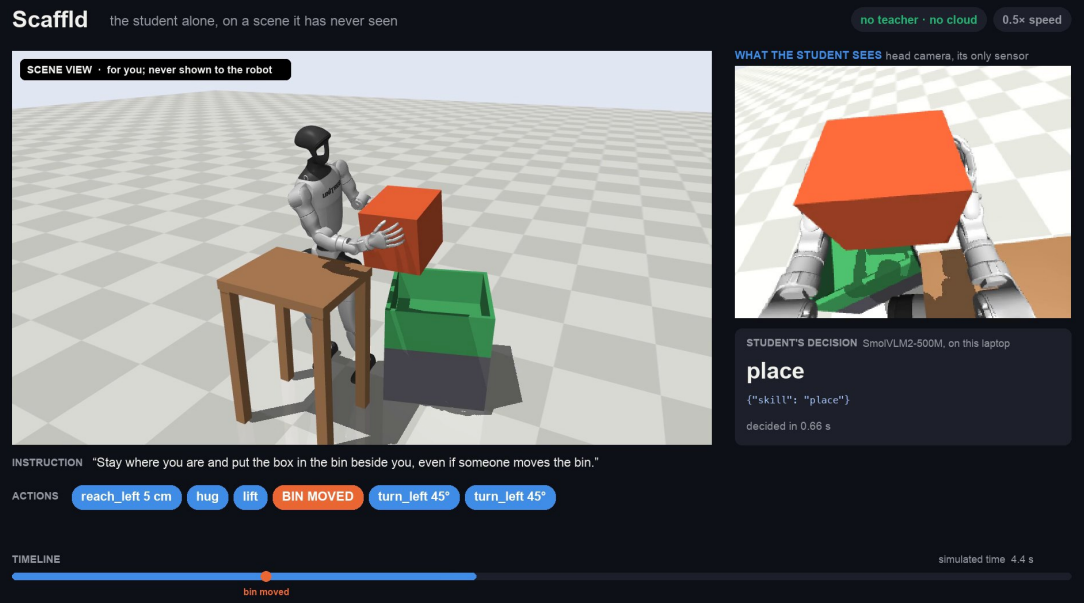
AgentVerse 2026 · Robotics Track

Scaffld

A robot teacher that finds the gaps in a robot's skills and builds its own lessons.

One typed sentence → a new robot skill in 39.7 minutes.

No human demonstrations. The trained student then runs alone on a laptop: no teacher, no cloud.



Owen Yan · Isaac · Santi · Anh · Josh Lim

github.com/owenbonc/isaac-s-mom

The problem

Robots learn from human demonstrations. That doesn't scale.

50 collectors × 12 months

to record 76k trajectories (DROID)

13 robots × 17 months

to collect 130k episodes (RT-1)

Every new task, and every new failure, costs **more operator hours**.

DROID: Khazatsky et al., arXiv:2403.12945. RT-1: Brohan et al., RSS 2023.

Our idea

Let a teacher agent create the lessons.

TEACHER · Argon, cloud

Writes the task from one sentence.

Demonstrates by driving the robot through its own camera and skills.

Diagnoses the student's failures and teaches the fix.

VALIDATOR · ground truth

Judges every demo by a declarative success rule.

Keeps only what works: unsafe or failed demos never train.

Says **why** a failure happened, in words.

STUDENT · 500M, laptop

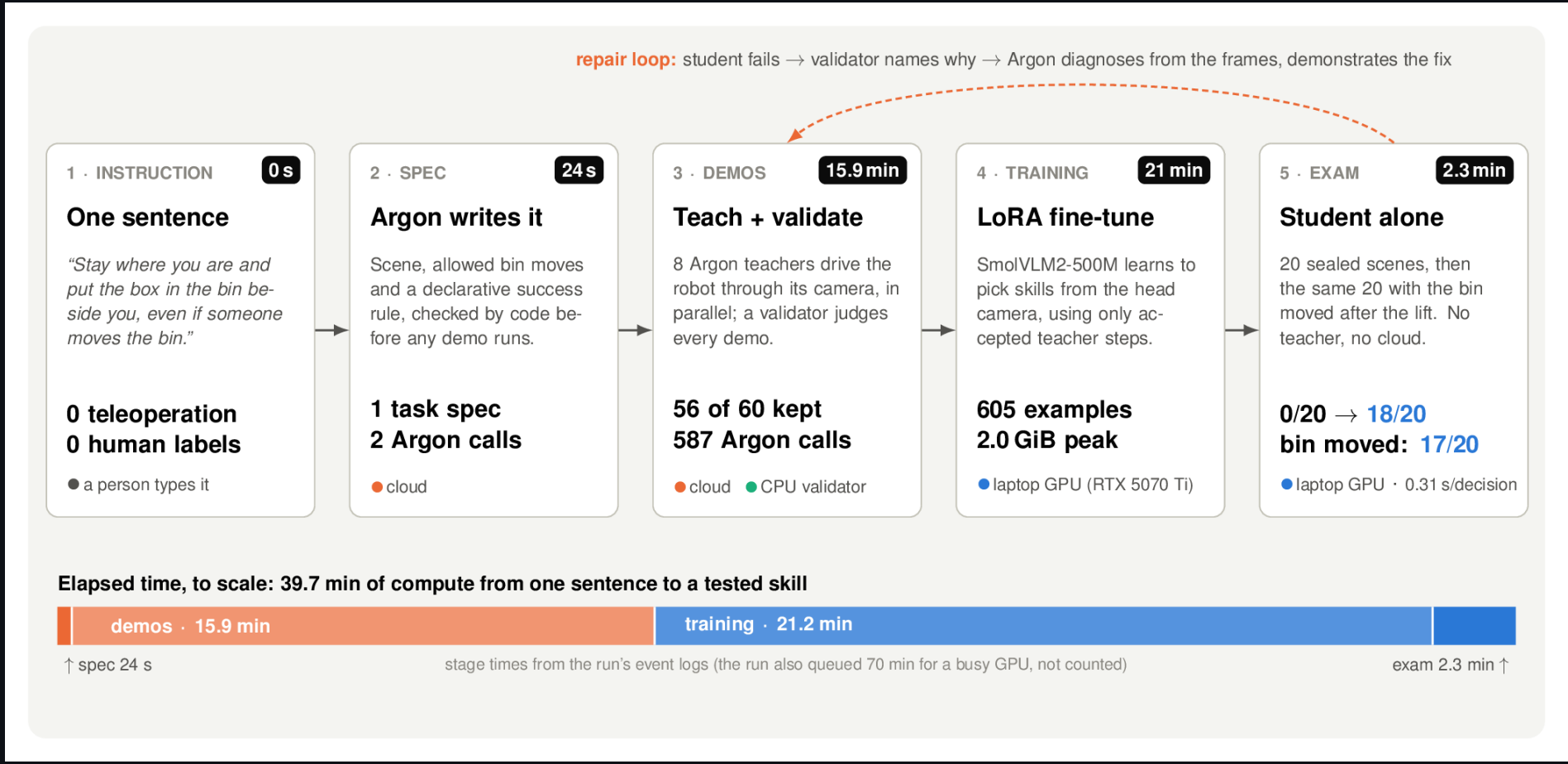
Learns from the accepted teacher steps (LoRA).

Sees **only the head camera**.

Runs alone: no teacher, no cloud, 0.31 s per decision.

How it works

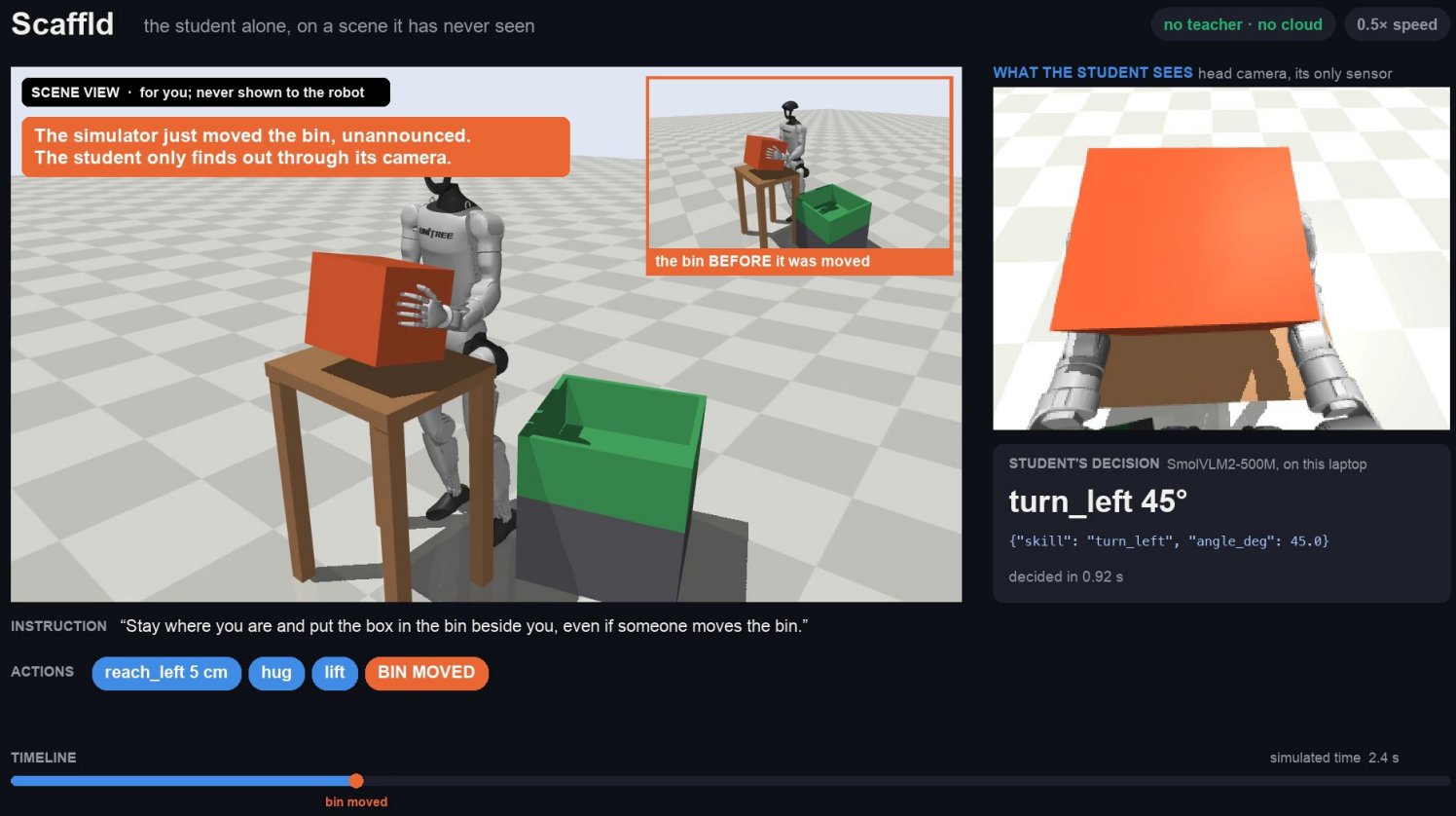
One sentence → a tested skill in 39.7 minutes



Stage times measured from the run's event logs; a 70-minute wait for a busy GPU is not counted.

Watch

The student alone. The bin moves mid-task.



One continuous take from the simulator · perception beside action · the simulator moves the bin, unannounced

Result 1

From 0/20 to 18/20on scenes it has never seen

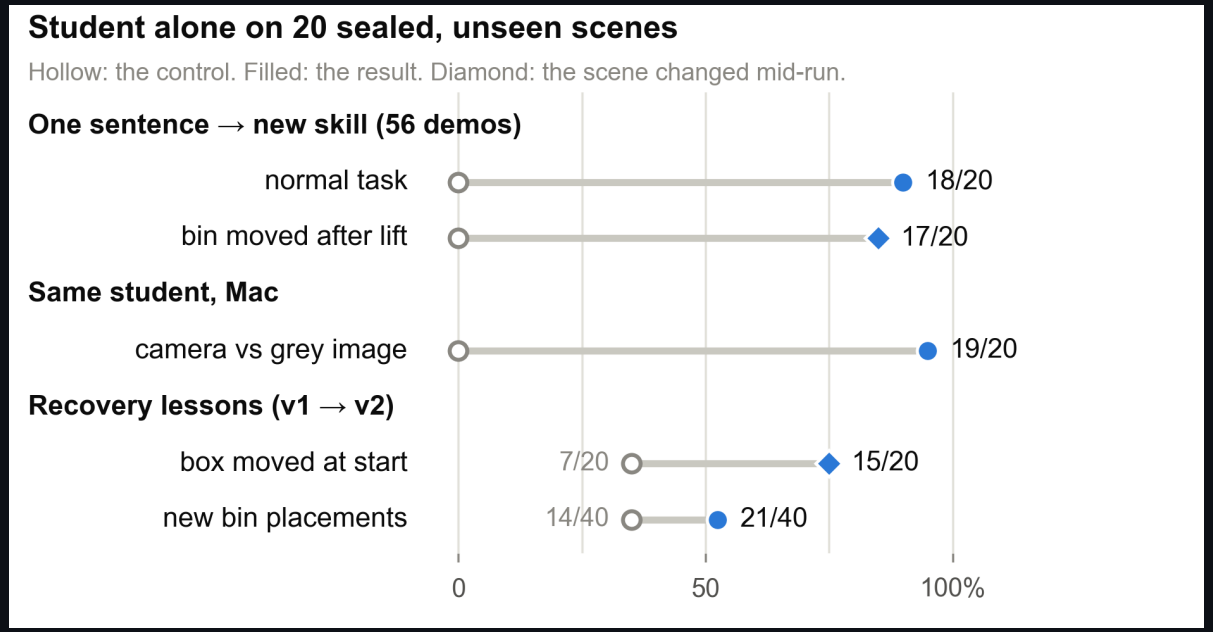
18/20

sealed, unseen scenes, student alone

17/20

with the bin moved after the lift

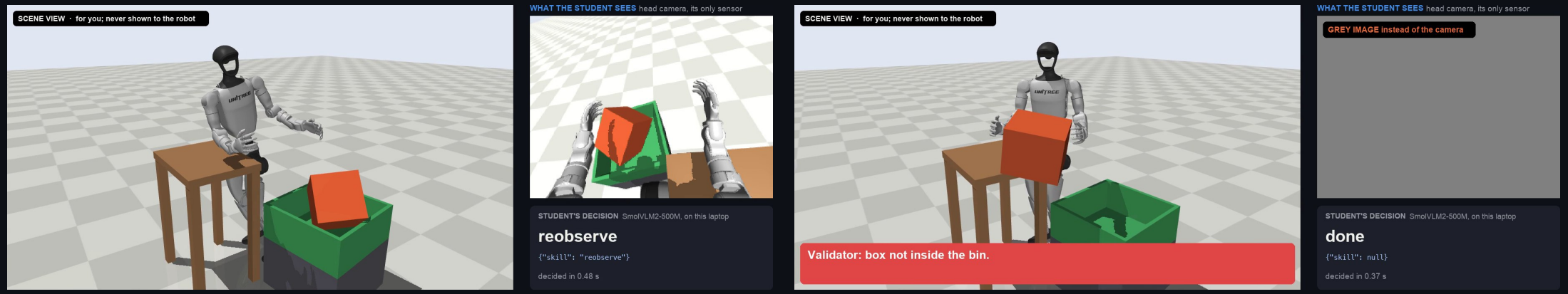
untrained model: 0/20 and 0/20
McNemar $p < 0.001$, paired by seed



Trained on 56 accepted teacher demos (605 examples, 21-minute fine-tune). Every count traces to a saved episode.

Result 2

It acts on what it sees.



camera: 19/20

It twists until the bin is under the box, then places.

grey image: 0/20 (Mac)

Same scene, same student: one twist, and it misses.

The demonstrations teach when to stop: trained on 14 or 28 demos it never places (0/20); on 56, **18/20**.

Camera vs grey image: same checkpoint, seeds and machine (Mac); only the image differs.

Result 3

Lessons aimed at failures work.

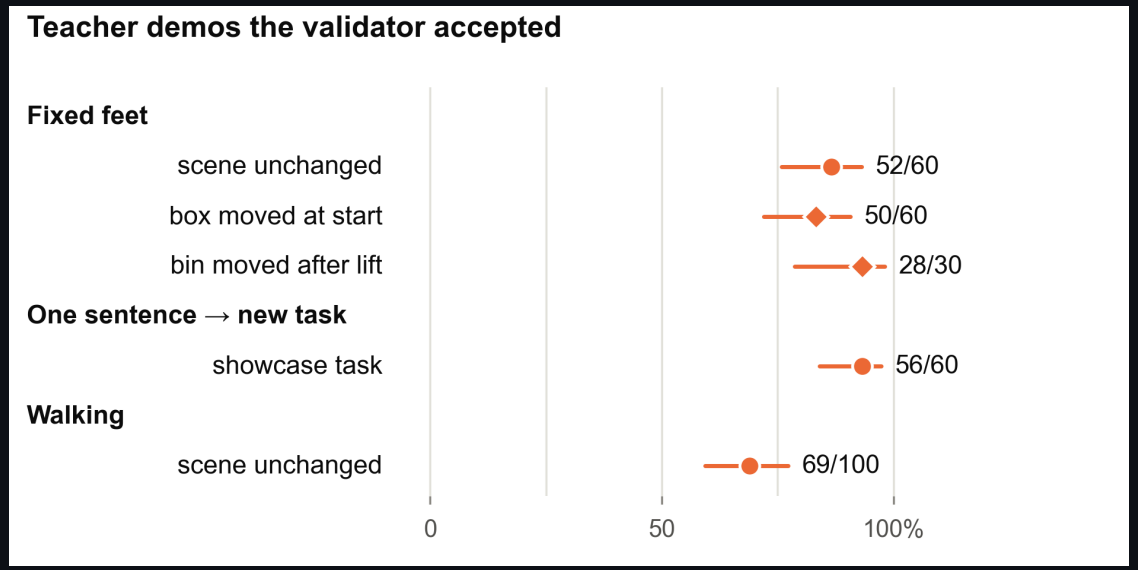
Box moved at the start

7 → 15 /20

Bin placements never seen in training

14 → 21 /40

overall $p = 0.19$; bin further ahead 4/20 → 14/20, $p = 0.006$



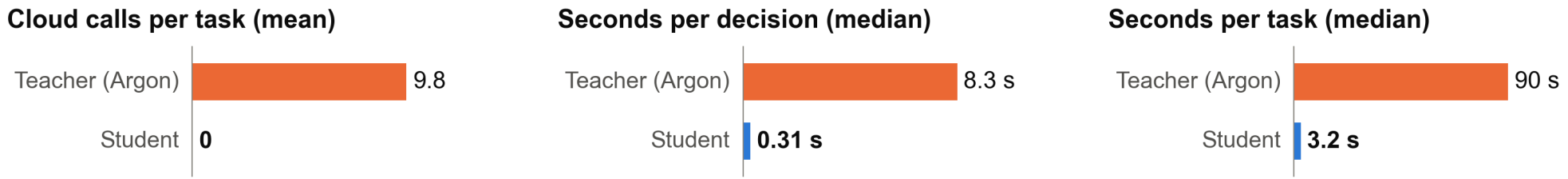
The teacher scales: the validator accepts 56/60 demos for a new task, and 69/100 while walking.

Adding recovery and box-moved demos (52 → 117) to the fixed-feet student; same seeds, student alone.

Impact

No humans in the loop. Nothing in the cloud at run time.

Doing the task: the cloud teacher vs. the student it trained



Teacher: 60 demos on development seeds (Mac). Student: 40 held-out episodes (laptop GPU). Success: teacher 56/60; student 18/20 and 17/20, alone.

1 sentence

human input: 0 teleoperated steps, 0 labels
DROID: 50 collectors × 12 months

20.9 GPU-min

training, on one laptop GPU
589 Argon calls for the whole skill

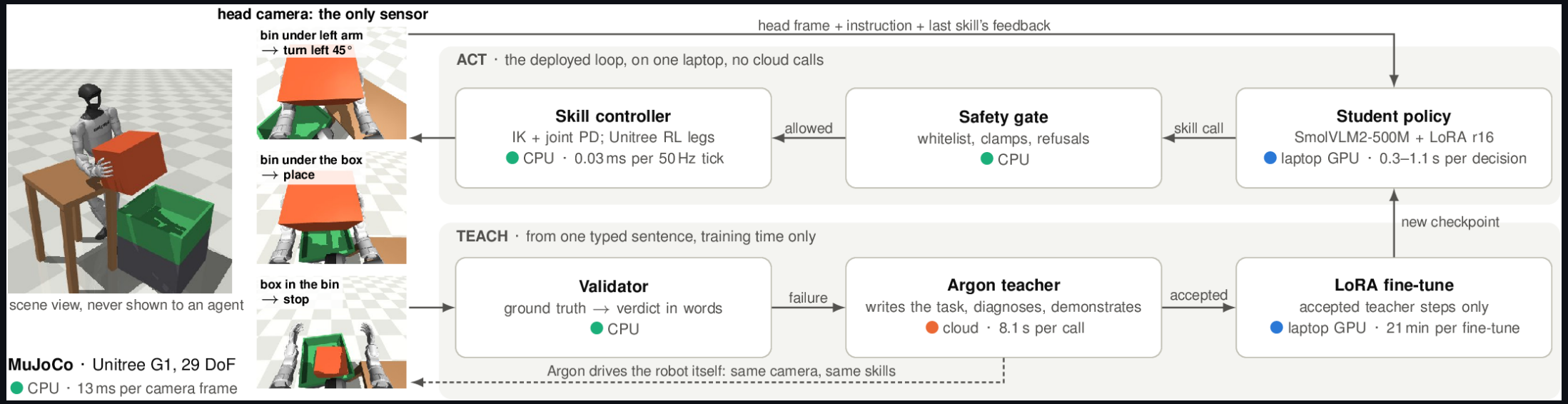
0.31 s

per decision, 0 cloud calls
teacher: 8.3 s and 9.8 calls per task

Teacher: 60 demos on development seeds. Student: 40 held-out episodes. Measured from event logs.

Architecture

Camera in, skills out. Ground truth only for the judge.



Student on the laptop GPU: 0.31 s per decision, **2.01 GiB of a 3 GB budget**.

Argon in the cloud, teaching only: 8.1 s per call. The deployed loop never calls it.

Safety and limits

Safe by construction. Honest about what's next.

SAFETY

No stale plans: every skill is chosen from the newest camera frame.

Refused before anything moves: past a joint limit, an unreachable target, a walk into the table.

Unsafe demos never train: the validator rejects falls, limit violations, penetration.

Sealed test seeds, an 800-call cap, keys never logged; **600+ tests**.

NEXT

Sim-to-real. Everything so far is in simulation.

Re-aiming the student. It keeps its learned routine when the bin moves beyond its training range; the teacher already re-aims.

Walking. The teacher walks (69/100 demos accepted); the student has its first successes (2/20).

Scaffld

Teach a robot a new skill in 39.7 minutes.

One sentence. No human demonstrations. A student that runs alone.

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